

Auditing a live packing benchmark: certified GPU refinement and the anatomy of polygon-packing records

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August 2026

Abstract

Erich Friedman’s Packing Center has curated best-known packings of regular polygons in regular polygons for more than twenty-five years, and in 2025–26 it became an arms race: industrial NLP-solver teams, DeepMind’s AlphaEvolve, and a wave of prolific individual contributors now improve its ~ 490 entries almost daily. We audit this benchmark with a certified pipeline — batched float32 GPU search, on-device float64 refinement, and independent exact separating-axis certification — and find that the modern record wave is systematically under-converged by contributor-specific amounts: entries by two of the most prolific setters sit 10^{-4} – 4×10^{-3} above the bottoms of their own basins, while other contributors, all legacy (1996–2012) entries, and all closed-form entries are converged to ties. Exploiting this gap and three structured construction mechanisms, we claimed improved packings for 54 problems in five days; 52 were credited and 41 still stand six weeks later. Two of the new records are exact: 160-digit contact-manifold solving with integer-relation detection proves 41 and 42 unit squares pack in equilateral triangles of side exactly $6 + 8/\sqrt{3}$, and 43 squares in side $5 + 10/\sqrt{3}$, closed forms three prior record iterations missed. Negative results delimit what compute alone can do: random multi-start reproduces known optima up to $n \approx 22$ but cannot reach rearrangement-class basins, locating the frontier in construction, not optimization. All packings, table snapshots, and a one-command verifier are public.

1 Introduction

Since the late 1990s, Erich Friedman’s *Packing Center* [1, 2] has maintained tables of the smallest known regular M -gon containing n non-overlapping unit regular m -gons — 24 categories, roughly 490 entries, each attributed to the person who found it. For most of its history the site was a quiet corner of recreational geometry. That changed in 2025–26. DeepMind’s AlphaEvolve used these tables as a public testbed for LLM-evolved search programs and took records that still stand [3]; a FICO/ZIB team showed that an off-the-shelf spatial branch-and-bound stack takes dozens of entries in one sweep [4, 5]; evolutionary follow-ups beat AlphaEvolve at its own entries [6]; and a handful of individuals began mass-producing records, several hundred between them, with undisclosed pipelines. During the five days of our campaign the tables moved daily.

This paper treats that moving benchmark as an object of study. We ask three questions. *How sound are the current records* — are the listed values converged, and would they survive independent certification? *Where do new records actually come from* — what does each mechanism (more compute, better refinement, better construction) actually yield? And *what does it take to claim a record correctly* on a benchmark that moves under the claimant?

Our instrument is a certified two-phase pipeline (§3): massively batched float32 search on a consumer GPU, float64 refinement of survivors, and an *independent* certifier that computes exact separating-axis margins and a rigorous size bound for every claim. On top of it we operate four structured mechanisms (§4): reconstructing competitors’ published images and converging them further; structured basin-hopping with vacancy and lattice-seeded moves; *drop-one propagation*, which turns one strong $(n+1)$ -packing into records at n and below; and exact contact-manifold solving with PSLQ integer-relation detection, which here both identifies exact closed forms and certifies their absence.

The audit (§5) produced 54 claimed improvements across 14 tables, of which 52 were credited and 41 still stand at the ledger freeze six weeks later (Table 1); every claim in this paper is backed by public coordinates, a fresh certification run, and archived snapshots of the tables (§7). We report the findings a trophy list would hide: the margin distribution by prior record holder (Fig. 2), which shows the 2026 wave is under-converged in a contributor-specific way; the negative results (Table 2), which show random multi-start stops producing new records around $n \approx 22$; and the attrition curve (Fig. 3), which quantifies how quickly records on a live benchmark decay — two of our claims were invalidated purely by the table moving between verification and processing.

2 Records, semantics, and what counts as a claim

An entry in category (m, M) at count n is a placement of n unit-side regular m -gons, pairwise non-overlapping, inside a regular M -gon of side s ; smaller s is better. The site displays values truncated to five decimals with a trailing plus: a listed $s = x^+$ means the incumbent holds some value in $[x, x + 10^{-5})$. Beating an entry therefore requires a *certified* value strictly below the displayed floor x ; matching digits is a tie, and during an active wave the floor itself must be re-read the day of submission.

We call a packing *certified* when an independent checker — separate code from the search — verifies exact pairwise separation margins (separating axis over both polygons’ edge normals, exact for convex shapes) and exact containment margins in float64, then converts any residual violation into a rigorous bound by shrinking the inner polygons and dilating the container. All 54 claims certify with dilation cost at most 10^{-11} ; re-running the certifier on the exact coordinates we emailed reproduces every claimed value to 10^{-9} (§7).

Certification is not pedantry; it is where records are won and lost. Our float32 search stage accepts “packings” whose hidden violations reach 3×10^{-4} — large enough to fake a record — and finite-difference squeezing stalls $\sim 2 \times 10^{-4}$ above true basin bottoms. Both artifacts produced false records in our own runs before the two-phase discipline was in place, and §5 gives evidence that loose refinement of this kind, not invalid packings, is the signature failure of the current record wave.

3 A certified two-phase pipeline

Search, float32, batched. The engine solves many restarts simultaneously as one GPU program (JAX [18]): a penalty formulation (squared pairwise overlap plus squared containment violation) minimized by a natively batched L-BFGS [19] — two-loop recursion and Armijo backtracking vectorized over instances with per-instance step masks — inside a shrink-and-anneal outer loop with kick rescue for stalled instances. A single RTX 4090 sustains ≈ 127 restarts per

minute at $n \approx 36$, about $25\times$ a 32-thread CPU node running the reference implementation [7], at a total rented-GPU cost for the campaign of under \$100.

Refine, float64, on device. Each wave’s top survivors are promoted to float64 on the GPU and driven through polish (fixed-size penalty minimization to machine precision), grow-to-feasible, and squeeze (backtracking container shrink while a machine-precision-valid packing survives). Only then does the independent certifier of §2 run, followed by a same-day re-read of the live table — the last gate before any claim is emailed.

4 Four mechanisms, measured

1. Reconstruct and squeeze (the audit instrument). Record pages publish images, not coordinates. We recover coordinates from a published GIF/PNG to $\sim 0.1\text{--}0.15$ px by fitting shape boundaries subpixel, then converge the reconstruction in float64. If the incumbent’s pipeline was loose, the same configuration certifies *below their own displayed floor*; if it was tight, we document a tie. This mechanism is simultaneously our audit probe and our most productive record source: 45 of 54 claims (44 of the 52 credited), with margins up to 3.9×10^{-3} against the quoted entry. It found closed forms too: reconstructing the freshly-set $n = 42$ and 43 squares-in-triangle entries and squeezing revealed values 4×10^{-9} above $6 + 8/\sqrt{3}$ and $5 + 10/\sqrt{3}$ — staircase-family forms the incumbent, and two record iterations before him, had approximated numerically without recognizing (Fig. 1, center; confirmed exact in mechanism 4).

2. Structured basin-hopping. Random multi-start converges whatever basin it lands in but almost never lands in a globally rearranged one (§5). Our basin-hopping layer adds what randomness lacks: an elite pool; vacancy, teleport, shear, and coherent tilt-cluster moves; and *lattice immigrants* — exact lattice packings with deliberate defects injected into the population. Its signature result is 36 squares in an octagon (Fig. 1, left). Our first submission (2.92902...) beat the listed entry and was accepted — then superseded within minutes when the site’s maintainer noticed a trivial 37-square diamond lattice at $10 - 5\sqrt{2} = 2.92893^+$ dominating both $n = 36$ and 37 (“you did have the record, for about 5 minutes”). Re-seeding the search with *lattice-minus-one* configurations found the vacancy rearrangement at $s = 2.9287015\dots$, reclaiming the record over the trivial construction. The episode is now a checklist item: verify no trivial $(n+1)$ -construction dominates before claiming n .

3. Drop-one propagation. Removing one shape from a strong $(n+1)$ -packing and re-squeezing all $n+1$ removal variants costs one batched GPU call, and when the $(n+1)$ basin has better architecture than the incumbent at n , the improvement cascades downward. From our own 35 triangles-in-octagon basin this produced $n = 34$ at 1.86454^+ (Fig. 1, right; 7.2×10^{-3} below our week-old value) and then $n = 33$ at 1.84789^+ , halting at $n = 32$ where the incumbent architecture is already strong. Measured yield across ~ 400 seed packings: exactly one such cascade — rare, but each hit is worth multiple records, and the mechanism explains the observed monotonicity anomalies on the site (at this writing the triangles-in-pentagon table still lists $n = 44$ at 3.52140^+ , worse than $n = 45$ at 3.51261^+ , an inconsistency drop-one would repair if the $n = 45$ coordinates were public).

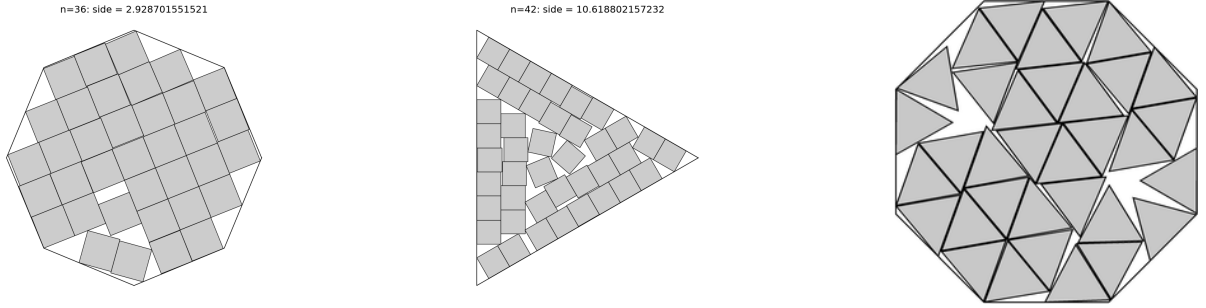


Figure 1: Flagship records of three mechanisms. *Left:* 36 squares in an octagon of side 2.92870^+ — a vacancy rearrangement of the trivial 37-square diamond lattice, found by basin-hopping seeded with lattice-minus-one configurations. *Center:* 42 squares in an equilateral triangle of side exactly $6 + 8/\sqrt{3}$, recovered from the incumbent’s published image and squeezed onto the closed form (proved exact in §4). *Right:* 34 triangles in an octagon of side 1.86454^+ , obtained by deleting one triangle from a stronger 35-triangle packing and re-squeezing.

4. Exact contact solving and PSLQ. For converged packings we extract the active contact graph (vertex–edge, vertex–wall), select independent constraints by pivoted QR, and solve the contact-manifold KKT system by minimum-norm Gauss–Newton in 160-digit arithmetic [17]; convergence is quadratic ($|F| : 10^{-8} \rightarrow 10^{-160}$ in five iterations) even on floppy manifolds. PSLQ [16] then hunts integer relations on the refined s . Both outcomes occur, and both matter:

Identification. The squares-in-triangle configurations at $n = 41, 42$ solve to the exact root of $3s^2 - 36s + 44$, i.e. $s = 6 + 8/\sqrt{3} = 10.6188021535\dots$, and $n = 43$ to the root of $3s^2 - 30s - 25$, i.e. $s = 5 + 10/\sqrt{3} = 10.7735026918\dots$, matching the algebraic forms to all 65 computed digits. Our submitted float64 records sit 3.7×10^{-9} and 7.0×10^{-9} above the exact bottoms (identical truncated display). Contact structure: at $n = 43$ only 25 of 43 squares are load-bearing, on a 13-dimensional floppy manifold.

Refutation. For 26 squares in an octagon (our record $2.52709\dots$), the same pipeline certifies at 65 digits that *no* minimal polynomial of degree ≤ 8 with coefficients up to 10^8 exists, and no relation in $\mathbb{Q}(\sqrt{2} + \sqrt{2})$ despite the packing’s 22.5° tilt structure — a no-closed-form certificate (4 rattlers, 8 floppy modes, 59 independent contacts). The same computation acts as a super-polisher, locating basin bottoms a few 10^{-9} below float64 values.

5 The audit

Scale and outcome. In five days (July 4–8, 2026) the pipeline produced certified improvements for 54 problems across 14 of the 24 tables (Table 1). Fifty-two were credited on the live site; at the ledger freeze on August 16, 41 still stand. The two uncredited claims fell to the benchmark itself, not to invalid packings: one was overtaken by a competitor’s better value posted the day the batch went out, and one targeted a floor that was stale by the time of processing — we do not count either. Every one of the 54 packings, plus a certified tie used to validate the submission channel, re-certifies today from the exact coordinates submitted.

Convergence demographics. Figure 2 plots each claim’s margin against the prior holder of the entry. The pattern is stark and contributor-specific. Entries taken from the two most prolific 2026 setters cluster at margins of 10^{-4} to 4×10^{-3} : their pipelines produce genuinely

Table 1: The campaign ledger, summarized per table (freeze 2026-08-16). “Credited” means the credit is verified in an archived snapshot of the live table; margins are lower bounds against the displayed floor of the entry beaten. The per-problem ledger with full-precision values, mechanisms, and per-claim status is in the repository (`LEDGER.md`).

Table	n range	Claimed	Credited	Standing	Margin range
Squares in octagons	26–40	9	9	4	1.5×10^{-5} – 3.9×10^{-3}
Triangles in pentagons	28–43	8	8	8	5.8×10^{-5} – 3.3×10^{-3}
Pentagons in hexagons	6–31	7	6	4	9.3×10^{-5} – 3.2×10^{-3}
Triangles in octagons	31–37	7	7	7	3.0×10^{-5} – 7.4×10^{-3}
Hexagons in pentagons	5–32	5	5	4	1.6×10^{-6} – 1.6×10^{-3}
Pentagons in octagons	28–31	4	3	1	2.4×10^{-4} – 1.1×10^{-3}
Squares in triangles	41–43	3	3	3	4.3×10^{-4} – 7.6×10^{-4}
Hexagons in hexagons	32–33	2	2	2	1.3×10^{-4} – 7.6×10^{-4}
Octagons in hexagons	6, 20	2	2	1	1.5×10^{-4} – 2.3×10^{-3}
Octagons in triangles	14, 18	2	2	2	8.5×10^{-4} – 1.6×10^{-3}
Squares in hexagons	30, 40	2	2	2	1.1×10^{-5} – 1.6×10^{-5}
Pentagons in squares	8	1	1	1	3.9×10^{-7}
Pentagons in triangles	26	1	1	1	7.4×10^{-4}
Triangles in hexagons	36	1	1	1	1.2×10^{-5}
Total		54	52	41	3.9×10^{-7} – 7.4×10^{-3}

new configurations whose final refinement stops three to four decimals short of the basin bottom. Entries by other active contributors yielded only at the 10^{-5} scale or below — three fell only to exhaustive search at the sixth decimal (3.9×10^{-7} in the extreme case) — and an entire further class never yielded at all: every legacy entry we attacked (1996–2012, including four triangles-in-hexagon basins re-converged to nine digits) and every closed-form entry proved to be an exact tie. The benchmark’s weakness is thus not wrong packings but *loose refinement at scale*, invisible under five-decimal truncation until someone re-converges the published images.

Where compute stops. Table 2 summarizes the campaign’s dead ends. The pipeline reproduces known optima essentially everywhere it lands ($n \lesssim 22$), and at the one control point where a better basin is known to exist — triangles-in-hexagons $n = 21$ — thousands of independent starts never find it. On fresh large- n tails the gap is structural, not numerical: at $n = 28$ triangles-in-hexagons our best wave landed 1.1×10^{-2} above the incumbent, a different architecture altogether. The conclusion, consistent with the methods used by the currently successful setters [3, 4, 6], is that further progress is construction-limited: the payoff is in seeded, structured moves — lattice defects, warm-started global solves, evolved move programs — not in more restarts. Consistent with this, one published ablation reports symmetry *constraints* making 539 of 750 instances worse [4]; seeding with symmetric configurations, by contrast, is what won our $n = 36$.

Benchmark velocity. Figure 3 shows the fate of the 52 credits. Within six weeks, eleven were re-taken by four competitors; earlier, four *finished* records were beaten by a competitor while they sat staged for thirty hours awaiting submission, and were never sent. This velocity has methodological consequences we adopted and recommend: verify against the live table the day of submission; archive dated snapshots (ours are in the repository and back every status in

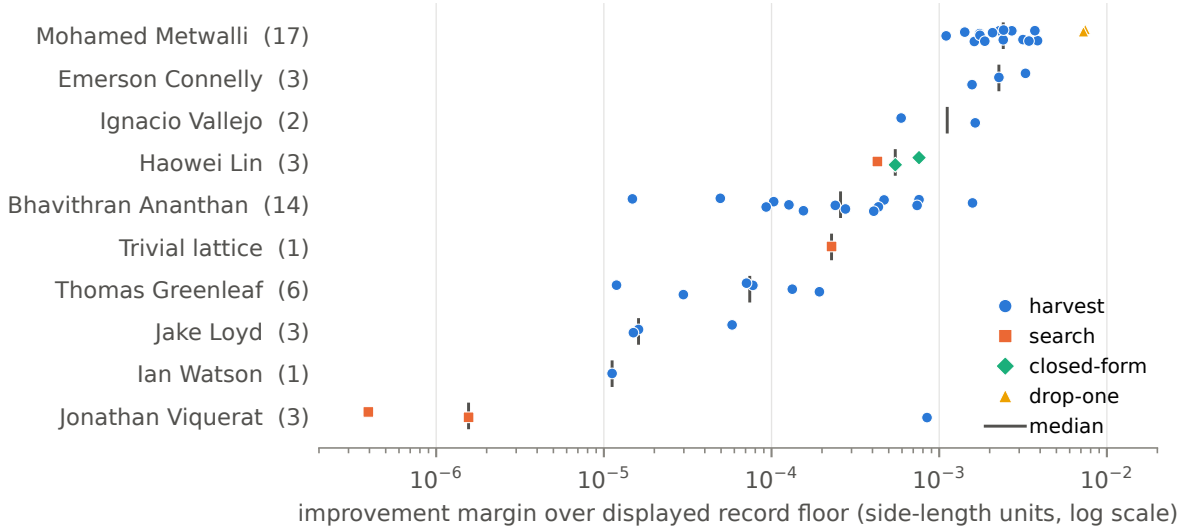


Figure 2: Improvement margin versus prior record holder for 53 claims (log scale; the stale-floor claim is excluded). Vertical bars mark per-holder medians; marker shape and color give the mechanism. The 2026 mass-production wave is under-converged by contributor-specific amounts — entries we took from the two most prolific setters sit 10^{-4} – 4×10^{-3} above their own basins’ bottoms — while other contributors’ entries yield only at the 10^{-5} – 10^{-6} scale, if at all. The margin is a lower bound on the incumbent’s convergence gap.

Table 1); and treat every claim as date-stamped. It also has a scientific one: a benchmark whose entries change daily but whose history is not archived anywhere silently loses the provenance of its own frontier.

6 Discussion

Three lessons generalize beyond this benchmark. First, *certification is cheap and decisive*: exact-margin verification plus a rigorous dilation bound costs milliseconds per packing, would have prevented every false record we generated internally, and — applied by a third party — revealed that a large fraction of a 490-entry benchmark sits measurably above its own basins’ bottoms. Live optimization benchmarks (packing sites, but equally leaderboards for heuristics) would benefit from requiring machine-checkable certificates and full-precision coordinates rather than images. Second, *refinement and construction are different resources*. GPU-scale refinement harvested 45 records here, but it is a one-time dividend: once the loose entries are gone (several competitors re-converged their own entries within days of our wave), what remains is the construction problem, where our thousands of restarts bought nothing that structure did not. Third, *exact solving belongs in the loop*: feeding converged numerical packings to arbitrary-precision contact solving identified three records as exact algebraic values ($6 + 8/\sqrt{3}$ twice, $5 + 10/\sqrt{3}$) and certified a fourth as having no simple closed form, at negligible cost next to the search that found them.

Limitations: all values are upper bounds — nothing here is proved optimal, in contrast to interval-arithmetic work on circle packings [9]; the audit covers one benchmark family (regular polygons in regular containers); and our exact-form statements certify the contact system of our

Table 2: Negative results. Each row is a documented dead end; together they locate the open frontier in construction rather than compute.

Experiment	Outcome
$n = 9$ – 12 , all 24 tables (69 entries searched)	Zero improvements; the band is fully converged by the 2025–26 wave, including AlphaEvolve’s hexagons-in-hexagons $n = 12$, which our searches tie but do not beat.
Legacy entries, e.g. triangles-in-hexagons $n = 8, 16, 17, 18$ (2005–2012)	Reproduced to 10^{-9} : exact ties. Legacy $\neq$ soft.
Triangles-in-hexagons $n = 21$ control (a better basin is <i>known</i> to exist)	Never found: 4,700+ independent starts plus 400+ structured basin-hopping rounds converge to 1.99877, above the known 1.99869 ⁺ — random multi-start cannot reach rearrangement-class basins.
Trivial plateaus (e.g. $s = 2$ at $n = 22$ – 24 triangles-in-hexagons; $4 + 4\sqrt{2}$ octagons-in-squares)	Rigid under every move class tried; float32 “sub-plateau” sightings were violations in disguise.
Drop-one sweep over ~ 400 seed packings	One cascade (§4); all other neighbors tight.

configurations, not global optimality at those n .

7 Reproducibility and disclosure

The toolkit, the per-problem ledger, all 54 coordinate sets (plus the validation tie), the certification table, and dated scrapes of all 24 record tables are public at <https://github.com/AriPapowitz/polygon-packing-records>. One command re-certifies every claim from the submitted coordinates (`paper/collect_and_certify.py`); the ledger and figures regenerate from the archived snapshots. The engine extends an open-source packer [7] (GPL-3.0) with the batched GPU search, float64 refinement, certification, reconstruction, and exact-solving layers.

This campaign was conducted with substantial AI assistance (Claude, Anthropic): the code was largely AI-written under the author’s direction, and overnight search campaigns ran autonomously. Every packing is machine-generated, independently certified by the tooling described here, and was verified by the site’s maintainer before listing.

Acknowledgments. Erich Friedman has maintained the Packing Center, and processed a great many submissions including ours, with patience and care. The competitors whose entries seeded our harvests — and who re-took eleven of them within weeks — made the benchmark the live, adversarial object this paper studies.

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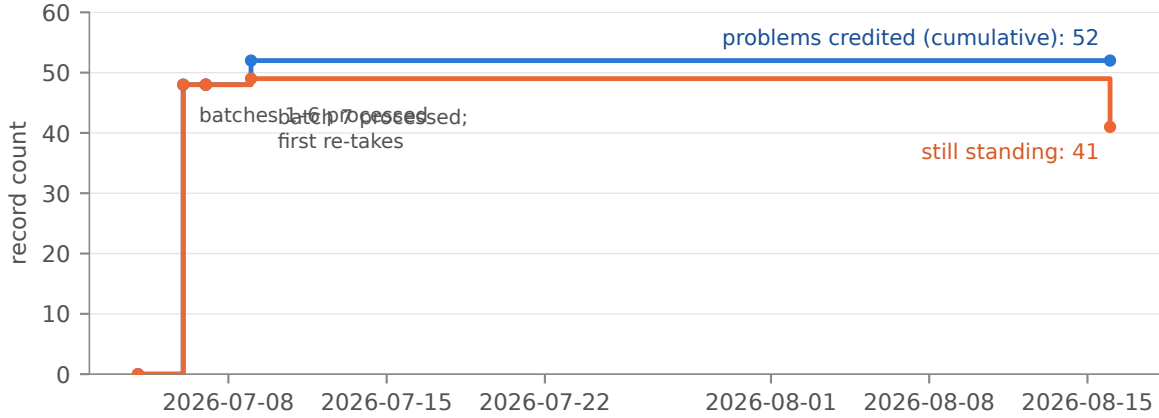


Figure 3: Benchmark velocity. Cumulative credited problems (blue) versus still standing (orange) across our archived snapshots. Eleven credits were re-taken within six weeks by four competitors; two further claims (not shown) were invalidated before processing by the table moving. Record claims on a live benchmark are date-stamped statements, not permanent facts.

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